

Measuring Matter

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Match the quantity to its tool and unit: **mass** → **balance** → grams (g); **liquid volume** → **graduated cylinder** → milliliters (mL); **length** → **ruler** or meter stick → centimeters (cm); **temperature** → **thermometer** → degrees Celsius (°C). Read a graduated cylinder at **eye level** at the **bottom of the curve**.

Round 1 — Retrieve It

1. Which tool measures mass, and what unit does it use?

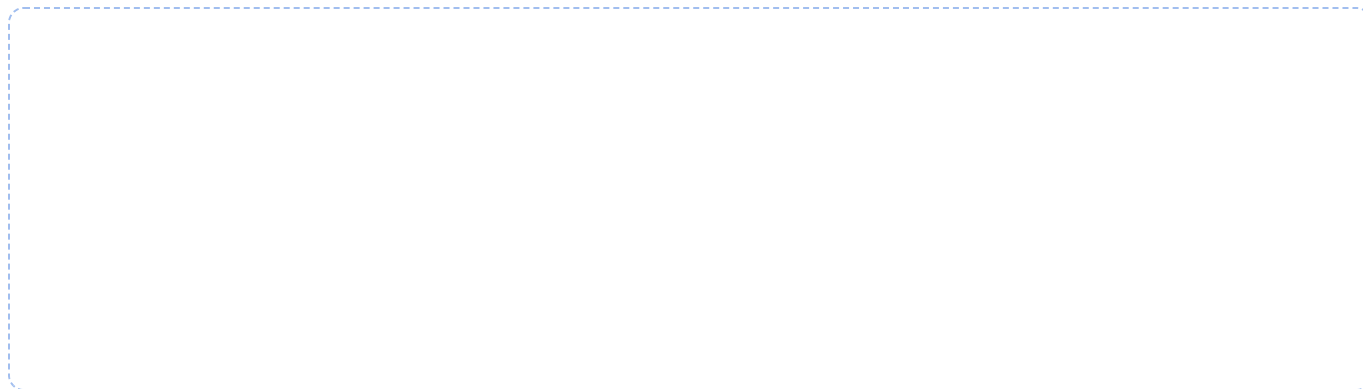
2. Which tool measures liquid volume, and what unit does it use?

3. Which tool measures temperature, and what unit does it use?

4. How should you look at a graduated cylinder to read it correctly?

Round 2 — Sketch It

Draw a graduated cylinder with water in it. Show the curved top of the water and mark where you would read the amount (the bottom of the curve, at eye level).



Measuring Matter — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Tool

1. You want to find the mass of a handful of marbles. Which tool and unit would you use?

2. You need to measure exactly 75 mL of juice. Which tool would you use, and how do you read it?

3. You want to know how tall your plant grew this week. Which tool and unit would you use?

Round 4 — Reason from Evidence

Two students measure the same water in a graduated cylinder. One reads it from above and gets 48 mL. The other reads it at eye level and gets 50 mL. Which reading should the class use, and why?

Measuring Matter — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain why each quantity needs its own tool and unit. Why can't a ruler measure liquid volume?

C**Connect**

Connect one measuring tool to a time you or someone at home used it (for example, cooking, checking a fever, or building). Name the unit.

E**Extend**

Plan how you would fully measure a bottle of water: its mass, its liquid volume, its height, and its temperature. List each tool and unit.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Measuring Matter

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. A balance; grams (g).
2. A graduated cylinder; milliliters (mL).
3. A thermometer; degrees Celsius (°C).
4. At eye level, reading the bottom of the water's curve.

Sketch It — look for

A graduated cylinder with a curved (dipping) water surface, and a mark or eye showing the reading taken at the bottom of the curve, at eye level.

Apply It

1. A balance, measured in grams (g).
2. A graduated cylinder; read at eye level at the bottom of the water's curve.
3. A ruler or meter stick, measured in centimeters (cm).

Reason from Evidence — look for

Use the eye-level reading (50 mL). Looking down from above makes the reading wrong; a graduated cylinder must be read straight on at the bottom of the curve for an accurate volume.

ACE — Extend look-for

Correct pairs: mass → balance → g, liquid volume → graduated cylinder → mL, length/height → ruler → cm, temperature → thermometer → °C.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.